

Chapter 1

Introduction

1.1 Introduction

1.1.1 Definitions

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j \in J} V^j$$

and we say T is a $|J|$ -legged tensor.

Given finite dimensional Vector spaces V and W , we have the identity that $V \otimes W^* \equiv \text{Hom}(W, V)$. Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V , $(V^*)^* = V$, we can peel off any number of V_i s, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \rightarrow V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t : V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

$$t = \sum_{i_1, i_2, j_1, j_2} T_{j_1 j_2 i_1 i_2} |v_{i_1}\rangle |v_{i_2}\rangle \langle w_{j_1}| \langle w_{j_2}|$$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \rightarrow (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.

1.1.2 Review of Symmetric Tensor networks.

Generally, we follow the notations of [SV12], [Sch+20], [Sch+20], [Sch20] which gave an overview for implementing $SU(2)$ invariant tensor networks. A general overview for (Non)abelian symmetries is [SPV10], and a detailed account for the abelian $U(1)$ in [SPV11]. There are a multitude of implementations given in major packages, such as ITensor, [FWS22], Cyten, TenPy, [Hau+24], QSpace, [Wei24] ChemTensor, UniTensor, YasTn (Abelian Only) [Ram+25] (MP-SKit?), (Cytex (Abelian only i think.)). A general account of symmetries is given in [Unf24] with its respective implementation in [Hau+24] (We have to build cyten citation manually.)

1.1.3 Quantum Gates as Tensor Network Operators

As a sanity check, view this in context of a quantum circuit. A quantum circuit is defined as (unitary) map $(\mathbb{C}^2)^n \rightarrow (\mathbb{C}^2)^m$.

$$CNOT : C^2 \otimes C^2 \rightarrow C^2 \otimes C^2 = \sum_{ij} |i, i \oplus j\rangle \langle i, j|$$

$$H = \frac{1}{\sqrt{2}} \sum_{ij} (-1)^{ij} |i\rangle \langle j|$$

1.2 Symmetries

What do we know now? First of all, we take a symmetry group G that is finite and compact (why?)[For compact groups, representations are completely reducible (Maschke) and the irreps are finite aswell. Finite groups are contained in compact groups.]. Then we look for its unitary representation $U(G)$ on a Hilbert space H . Furthermore, if this is a multiparty hilbert space, lets say n sites, we view the unitary representations U_n at site n . The unitary representation $U_n : G \rightarrow Hom(V_n, V_n)$ is a covariant linear unitary map. A map $V_i \rightarrow V_j$ is symmetric if it is an intertwiner of the representations, i.e. the following diagram commutes:

$$\begin{array}{ccc} V_i & \xrightarrow{U_{V_i}(g)} & V_i \\ f \downarrow & & \downarrow f \\ V_j & \xrightarrow{U_{V_j}(g)} & V_j \end{array}$$

Note that here V_i is called a *symmetry space*.

Some examples of these groups include $U(1)$, $SU(2)$, $\mathbb{Z}_n$

Now where the intuition is more shaky: Consider a symmetry space V_i on a single site (why single). Then we get that it decomposes into a direct sum of irreps V_i^a ,

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a$$

and we can easily construct an isomorphism (but this depends on choosing a basis for D_a .)

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a = \bigoplus_a (D_a \otimes V_i^a).$$

Well, the isom we can construct here would look like $\bigoplus_1^{d_a} V_i^a \rightarrow D_a \otimes V_i^a$.

Consider a single site of 3 spin 1/2 particles. Under $SU(2)$ this decomposes as

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$$

Now consider that C^2 is just the fundamental representation of $SU(2)$.

Note here, an invariant subspace $W \subseteq V$ is a subset of vectors s.t. for all $U \in SU(2)$, $Uw \in W$.

symmetry obviously implies pullthrough properties, do Intertwiners as well?

Yes, because $U(V \otimes W) : g \rightarrow U_V(g) \otimes U_W(g)$ and the following diagram commutes:

$$\begin{array}{ccc} V_i \otimes V_j & \xrightarrow{U_{V_i \otimes V_j}(g)} & V_i \otimes V_j \\ T \downarrow & & \downarrow T \\ V_k & \xrightarrow{U_{V_k}(g)} & V_k \end{array}$$

1.2.1 Some Symmetry Groups

$\mathbb{Z}_n$

This group is given by integers $0, \dots, n-1$ with addition mod n . Since this happens to be abelian, all the irreps are 1 dimensional, i.e. $U_a(g) : g \rightarrow \text{Hom}(\mathbb{C}, \mathbb{C})$ with representatives $U_a(g) = z \mapsto e^{2\pi i a g / N} z$. What is the fusion rule here?

$$a \otimes b \equiv (c := a + b \mod n)$$

This also does not have fusion multiplicities.

The group $SU(2)$

Summarize the results from the paper here.

The group $SU(2)_3$

Technically this is a fusion category, the thesis [Unf24] handles this distinction. But if we think of this in terms of the programming guide, this seems to be doable using a group.

Now, since we can argue that Products of symmetry groups again form a symmetry group (proof?), we would also need another type for sector.

We could use `classSector(NamedTuple)` holding an array charges.

1.2.2 Notes on structural tensors and fusion trees

What do the CG tensors look like? As they are three index and have no intermediates, these are simply $Q_{j_1, j_2, j_3} = (C^{fuse})_{(j_1, m_{j_1}), (j_2, m_{j_2})}^{(j_3, m_{j_3})}$.

1.2.3 Notes on the papers

Lets assume we only look at incoming legs, i.e. we look at T_{j_a, j_b, j_c} in a composite vectors space $V^a \otimes V^b \otimes V^c$. By the decomposition

$$V^a \otimes V^b \otimes V^c = \left(\bigoplus_{j_a} D_{j_a} \otimes V_{j_a} \right) \otimes \left(\bigoplus_{j_b} D_{j_b} \otimes V_{j_b} \right) \otimes \left(\bigoplus_{j_c} D_{j_c} \otimes V_{j_c} \right)$$

and we can represent a single index inside the block space

$$T_{j_a, j_b, j_c} \in (D_{j_a j_b j_c} \otimes V_{j_a j_b j_c})$$

Choosning a basis $t = (t_{j_a}, t_{j_b}, t_{j_c})$ and $m = (m_{j_a}, m_{j_b}, m_{j_c})$ we can write

$$T_{j_a, j_b, j_c} = \sum_{t, n} (T_{j_a, j_b, j_c})_{t, m} (|t\rangle \otimes |m\rangle)$$

Now we can use Wigner-Eckart to obtain

$$\begin{aligned} T_{j_a, j_b, j_c} &= \sum_{t, n} ((P_{j_a j_b j_c})_t \cdot (Q_{j_a j_b j_c})_m)_{t, m} (|t\rangle \otimes |m\rangle) \\ &= \left(\sum_t (P_{j_a j_b j_c})_t |t\rangle \right) \otimes \left(\sum_m (Q_{j_a j_b j_c})_m |m\rangle \right) = P_{j_a j_b j_c} \otimes Q_{j_a j_b j_c} \end{aligned}$$

If we generalize this with the decomposition for k indices $i = (i_1, i_2, \dots, i_k)$ and let $j_e = (j_{e_1}, j_{e_2}, \dots, j_{e_{k-3}})$ be the internals

$$T_i = \sum_{j_e} (P_j^{j_e})_t \cdot (Q_j^{j_e})_m$$

1.2.4 Notes on the decomposition

Suppose that (V, ρ_V) and (W, ρ_W) are representations of G .

Definition 1.2.1. A linear map $f : V \rightarrow W$ is called an *intertwiner* if

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad g \in G.$$

Define the *Vector space of Interwtiners* as

$$\text{Hom}_G(V, W)$$

Suppose now V^a is one of the reps that decomposes as

$$V^a \simeq \bigoplus_{i=1}^{d_a} R_a.$$

Lemma 1.2.2. For a vector space W , there is an isomorphism

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{ii \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \rightarrow \bigoplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \dots, w, \dots, 0)$$

and linear, thus

$$\Phi \left(\sum_i c_i e_i \otimes w_i \right) = (c_1 w_1, \dots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. $\square$

Lemma 1.2.3 (Isotopic Decomposition). Let C be a semisimple category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition

$$V \cong \bigoplus_a \text{Hom}_C(R_a, V) \otimes R_a. \quad (1.1)$$

where we refer to $\text{Hom}_C(R_a, V)$ as the *degeneracy* of the space.

TODO change this so its a vector space of dimension of the degeneracy, but discuss how to formulate this.

Proof. Since C is semisimple, every object decomposes as

$$V \equiv \bigoplus_a m_a R_a$$

for the degeneracies $m_a \in N$ (this is technically abuse of notation, if we define the direct sum.)

Applying $\text{Hom}(R_a, -)$ (this is a linear functor?) yields

$$\text{Hom}(R_b, V) = \bigoplus_a \text{Hom}(R_b, m_a R_a) = \text{Hom}_C(R_b, R_a)^{\oplus m_a}$$

and then by Schur's lemma

$$\text{Hom}_C(R_a, R_b) = \begin{cases} \mathbb{C}, & \text{if } a \equiv b; \\ 0 & \text{otherwise} \end{cases}$$

and therefore $\dim \text{Hom}_C(R_a, V) = \dim \mathbb{C}^{m_a} = m_a$.

An issue here might be that $\text{Hom}_C(R_a, V)$ is not an object or a morphism of $\text{Rep}(G)$. Its elements are, but not itself, it is in Vect . $\square$

So we actually need this: Maybe this is simpler if we always let the degeneracy be C ?

Lemma 1.2.4. Let C be a k -linear category and let D, E be finite-dimensional k -vector spaces. For any objects $R, S \in C$, there is a natural isomorphism

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. By unfolding the definitions and utilizing the properties of finite biproducts, we establish the result through a direct chain of natural isomorphisms:

$$\begin{aligned} \text{Hom}_C(D \otimes R, E \otimes S) &\cong \text{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by 1.2.2} \\ &\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \text{Hom}_C(R, S) && \text{by additivity of Hom}_C \\ &\cong k^{mn} \otimes \text{Hom}_C(R, S) && \text{by 1.2.2} \\ &\cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S) && \text{since } \text{Hom}_k(D, E) \cong k^{mn} \end{aligned}$$

$\square$

Theorem 1.2.5 (Decomposition of Semisimple Categories). Let C be a semisimple k -linear category with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of objects in C . Every morphism

$$T \in \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_i W_i \right)$$

admits a decomposition

$$T = \bigoplus_{\hat{a}, \hat{b}} \sum_{\mu} P^{(\hat{a}, \hat{b}, \mu)} \otimes Q^{(\hat{a}, \hat{b}, \mu)},$$

where

$$P^{(\hat{a}, \hat{b}, \mu)} \in \text{Hom} \left(\bigotimes_i D_{i, a_i}, \bigotimes_j E_{j, b_j} \right),$$

and

$$Q^{(\hat{a}, \hat{b}, \mu)} \in \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}).$$

Proof. Since C is semisimple, every object decomposes as a finite direct sum of simple objects. Thus, by Lemma ?? and the distributivity of the tensor product through the biproduct

$$\begin{aligned} \bigotimes_i V_i &\cong \bigotimes_i \left(\bigoplus_{a_i} D_{i, a_i} \otimes R_{a_i} \right) \\ &\cong \bigoplus_{\hat{a}} \left(\bigotimes_i D_{i, a_i} \right) \otimes \left(\bigotimes_i R_{a_i} \right). \\ &\cong \bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}} \end{aligned} \tag{1.2}$$

Then

$$\begin{aligned} \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_j W_j \right) &\cong \text{Hom}_C \left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}} \right) \quad \text{by (1.2)} \\ &\cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) \quad \text{by additivity of } \text{Hom}_C \\ &\cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}). \quad \text{by (1.2.4)} \end{aligned}$$

Choosing a basis

$$\{Q^{(\hat{a}, \hat{b}, \mu)}\}_{\mu}$$

of each space

$$\text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

every morphism T can therefore be written uniquely as

$$T = \bigoplus_{\hat{a}, \hat{b}} \sum_{\mu} P^{(\hat{a}, \hat{b}, \mu)} \otimes Q^{(\hat{a}, \hat{b}, \mu)},$$

with

$$P^{(\hat{a}, \hat{b}, \mu)} \in \text{Hom}(D_{\hat{a}}, E_{\hat{b}}).$$

□

example 1. If we restrict to the category $\mathbf{Rep}(SU(2))$ we can build the symmetric tensor networks as given in [SV12]. TODO: actual ly do this and convince yourself that the notations amd the triplet basis workd as desired.

1.2.5 Q intertwiners

Lets look at 2 paths for the 4 index tensor, $(a, b, c \rightarrow d)$ and $(a, b \rightarrow c, d)$.

$$(Q_{j_a j_b j_c}^{j_e})_{m_1, m_2, m_3, m_4} = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e; j_3 m_3 | j_4 m_4 \rangle$$

while the other topology would look like

$$Q \dots = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e | j_3 m_3; j_4 m_4 \rangle$$

and as an operator this would have to live in $\text{Hom}(V_{j_1} \otimes V_{j_2}, V_{j_3} \otimes V_{j_4})$

The yoga tree $(-1, 1, -3), (-2, 1, -4)$ Gives us a good mixture of those. This Q would then have to look like

$$Q_{-1-2-3-4}^1 = \sum_1 \langle j_{-1} m_{-1}; j_1 m_1 | j_{-3} m_{-3} \rangle \langle j_{-2} m_{-2} | j_1 m_1; j_{-4} m_{-4} \rangle = \sum_{m_1} C_{-1,1 \rightarrow -3} \cdot C_{-2,1 \rightarrow -4}$$

How would we do this when we have a contraction of two simultaneous indices:

$$Q_{-1-2}^{1,2} = \sum_{1,2} \langle -1 | 1; 2 \rangle \langle 1; 2 | -2 \rangle$$

If we define τ to be the fusion trees topology, then we can define ${}_{\tau}Q_j^i$

What would this look like for general symmetry groups G ? Lets choose a basis of its invariant decomp $|\lambda_a t_a \mu_a\rangle$. Then the generalized CG coefficient would look like

$$(C_{\lambda_a \lambda_b \rightarrow \lambda_c}^{\alpha})_{\mu_a \mu_b}^{\mu_c} = \langle \lambda_a \mu_a; \lambda_b \mu_b | \lambda_c \mu_c \rangle_{\alpha}$$

where alpha denotes the singular fusion multiplicity at μ_c .

Now to fully specify Q here for k sectors (λ_i, μ_i) , we need to give it $k - 3$ internal edges $(\lambda_{e_1}, \dots, \lambda_{e_{k-3}})$ along with $k - 2$ mulitplicity symbols, $(\alpha_1, \dots, \alpha_{k-2})$

Can we write down a general formula for Q ?

This would then be

$$Q_{\lambda}$$

1.3 Categories

1.3.1 Tensor and Fusion Categories

Let $\mathcal{C}$ be a category with bifunctor

$$\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}, \quad (X, Y) \mapsto X \otimes Y.$$

and the isomorphism $F : (- \otimes -) \otimes - \simeq - \otimes (- \otimes -)$

... introduce fusion categories here, with associativity and commutativity restrictions, I think this would be a good start.

Definition 1.3.1 (Monoidal Category). A monoidal category is a category $\mathcal{C}$ with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$

Let $\mathcal{C}$ and $\mathcal{D}$ be linear abelian categories. Then the *Deligne Tensor Product* $\mathcal{C} \boxtimes \mathcal{D}$ is a linear abelian category with a bifunctor

$$\boxtimes : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \boxtimes \mathcal{D}$$

with morphisms given by

$$\text{Hom}_{\mathcal{C} \boxtimes \mathcal{D}}(X \boxtimes Y, W \boxtimes V) = \text{Hom}_{\mathcal{C}}(X, W) \otimes \text{Hom}_{\mathcal{D}}(Y, V)$$

Definition 1.3.2 (The Deligne Tensor Product of Categories). Let k be a field, and let $\mathcal{C}$ and $\mathcal{D}$ be two k -linear, finite abelian categories. The **Deligne tensor product**, denoted by $\mathcal{C} \boxtimes \mathcal{D}$, is a k -linear finite abelian category equipped with a k -bilinear functor

$$\boxtimes : \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{C} \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 1.3.3 (The Category $\mathbf{Rep}_k(G)$). Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\mathbf{Rep}_k(G)$ are pairs (V, ρ_V) , where V is a k -vector space and $\rho_V : G \rightarrow \text{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\mathbf{Rep}_k(G)$, a morphism from V to W is a k -linear transformation $f : V \rightarrow W$ that is G -invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho_V(g)} & V \\ f \downarrow & & \downarrow f \\ W & \xrightarrow{\rho_W(g)} & W \end{array}$$

In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

The set of all such morphisms is denoted by $\text{Hom}_G(V, W)$.

We can further endow $\text{Rep}_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\mathbf{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Theorem 1.3.4. Let G_1 and G_2 be finite groups and let $\mathbf{Rep}(G_1), \mathbf{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories

$$\mathbf{Rep}(G_1 \times G_2) \simeq \mathbf{Rep}(G_1) \boxtimes \mathbf{Rep}(G_2)$$

Big issue here: $\mathbf{Rep}(\text{SU}(2))$ is not a fusion category as it is not finite. But if i read it correctly, the deligne product only requires abelianness of its groups. Now, $\mathbf{Rep}(G)$ is monoidal for any G but is it abelian?

So this is actually true, it is abelian. We could obtain a restricted versions by for example always restricting $\text{SU}(2)$ to some $\text{SU}(2)_k$ for some finite k . But i do not yet know how to present this nicely.

A second advantage would be that we can always represent fusion trees as monoidal categories (is this true or is fusion required here again?), and we can write down a unified theory for the fusion trees. Then I hope to write a nice method to view the "eval" of such a fusion tree, i.e. its clebsch gordan tensors by contracting down the topology of the tree.

1.4 Categories Needed

Definition 1.4.1 (Monoidal Category). A monoidal category is a quintuplet $(C, \otimes, 0, 1, \alpha) \dots$

Definition 1.4.2 (Semisimple Category). An abelian category C is semisimple if every object is either simple or a direct sum of simple objects. An object is *simple* if its only subobjects are 0 or X itself. Any object can be written as

$$X = \bigoplus_{i \in I} S_i$$

Lemma 1.4.3 (Schurs 1st lemma (which one?)). If S and T are simple objects in a semisimple category (over an algebraically closed field? what does this entail?), then it holds that

$$\text{Hom}(S, T) = \begin{cases} k, & \text{if } S = T; \\ 0, & \text{if } S \neq T. \end{cases}$$

Lets give a coordinatized approach to fusion categories:

Definition 1.4.4 (Fusion Category). A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_e^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

1.5 implementation

We want to be able to represent a traditional tensor op such as

```
1 np.einsum("ijklmn,nkol -> ijmo", a, b)
```

for symmetric tensors. As we want to reuse code form existing librariries that is written using numpy, it would be ideal to write our classes in a way that would only require substituting *np* for our own libraries *snp*. Towards this goal we define a abstractclass *Tensor*

```
1 class Tensor(abc.ABC):
2     ...
3     @abstractmethod
4     def transpose(cls, new_shape:list[int]): ...
5     ...
```

and we implement both an *np* wrapper *NPTensor* and our symmetric Tensor as *STensor*, i.e.

```
1 class STensor(Tensor):
2
3     def __init__(self, ...):
4         self.open_legs:list[int|str]
5         self.fusion_tree:FusionTree(self.open_legs, ...)
6         ...
7     @abstractmethod
8     def transpose(cls, new_shape:list[int]): ...
9     ...
```


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